

LESIBA LUCKY MANALA

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PROFESSIONAL SUMMARY

Final-year Computer Systems Engineering student at Tshwane University of Technology with hands-on experience in software development, industrial automation, embedded systems and software engineering. I enjoy designing solutions that combine hardware and software, from PLC controlled automation systems and Microcontroller-based IoT projects, to C++ applications and software development. I am passionate about solving engineering problems through practical, reliable and well-integrated systems.

TECHNICAL SKILLS

Software Development & Web technologies

Frontend	Backend & Database
HTML, CSS, Type Script	Node.js/Express
Next.js/React	PostgreSQL
Django	Oracle SQL in Database Design
Vercel	API Development and testing

- **Tools:** MS Office Suite (Excel data analysis, technical reporting), Git/GitHub.
- **Agentic tools:** Codex, Claude, Render, Anti-Gravity, Obsidian.
- **3D Modelling:** Unity 3D/ML-Agents, procedural terrain generation (Diamond Square algorithm)

Embedded Systems & IoT

- Microcontroller platforms: ESP32, Arduino Uno, FPGAs
- Real-time systems: ultrasonic distance measurement, water level monitoring, temperature sensing with ADC signal processing
- Communication protocols: Bluetooth, serial interfaces, wireless sensor networks

Industrial Automation & Control

- PLC Programming (Ladder Logic), SCADA/HMI dashboard development
- Sensor interfacing, I/O configuration, PWM/ADC integration
- System commissioning, hardware troubleshooting, circuit design

Networking & Infrastructure

- Network configuration, troubleshooting, CCNA-level routing and switching concepts

PROJECTS PORTFOLIO.

Industrial Automation

- **PLC Conveyor Belt System** - Designed and programmed a PLC-controlled sorting system that automatically separated metals from plastic materials using sensor feedback, eliminating manual sorting during testing and demonstrating reliable real-time automation

IoT & Embedded Systems

- **Water Level Monitoring System** – Developed an ESP32 based water monitoring system that provides real time tank level reporting, enabling users to monitor water levels without physical inspection.
- **Bluetooth Access Control** - Developed a secure entry system combining Bluetooth communication with microcontroller-based validation protocols.

- **Temperature Monitor** - Built an Arduino based temperature monitoring system that processes analogue sensor readings into accurate digital temperature measurements displayed in real-time, demonstrating accurate analog to digital signal processing and display integration.

Software Development

- **Unity ML-Agents** - Trained intelligent agents using reinforcement learning for autonomous task performance within simulated 3D environments.
- **Transportation software** - Currently developing a software that centralizes the informal sector of long-distance travel with private vehicles, bringing that side of the industry to light.
- **South African illness Tracker** - Developed a web application that tracks common and fatal illnesses in south Africa, bringing information that is otherwise out of reach to user's fingertips. <https://illness-tracker.vercel.app>
- **Portfolio** – Built a portfolio website to showcase all my skills and relevant projects I've encountered though out my educational journey. <https://my-portfolio-sepia-three-86.vercel.app>

EDUCATION

Diploma in Computer Systems Engineering | Tshwane University of Technology | 2023–Present

Certifications & Programs:

- **Cisco Networking Academy** (2023) – CCNA Modules, Networking Essentials
- **Tshimologong Digital Skills Development Program** (2022)
- **National Senior Certificate** (2022)

EXTRACURRICULAR ACTIVITIES

- **4IR Research Lab (2026)** – Volunteering at the Tshwane University of Technology Research laboratory as a software development Work Integrated Student to help alongside current research projects.
- **IBM Z Datathon (2024)** - Collaborated in a competitive team environment to develop data-driven solutions for real-world problems, strengthening analytical thinking and deadline-driven delivery.
- **Top Junior Leadership Program (2023)** - Completed team building, problem-solving workshops, and leadership skills development.

CORE STRENGTHS

- **Systems Thinking** – I can translate between hardware constraints and software requirements, ensuring integrated solutions that work end-to-end.
 - **Analytical Problem-Solving** - Strong debugging and system testing capabilities with a methodical approach to root-cause analysis.
 - **Collaboration** - Effective communicator who thrives in cross-functional teams, from academic project groups to competitive hackathons like the IBZ Datathon.
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